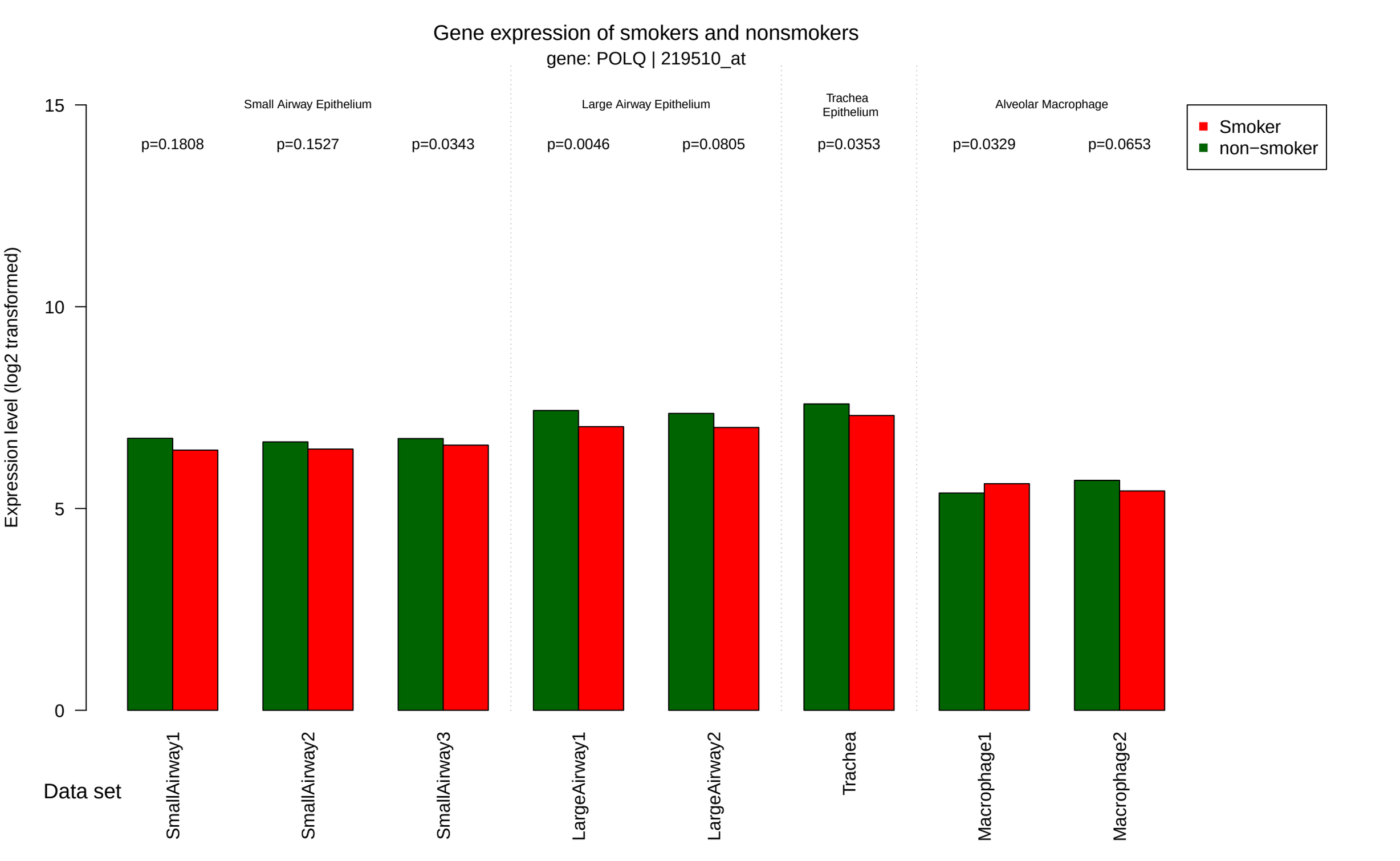
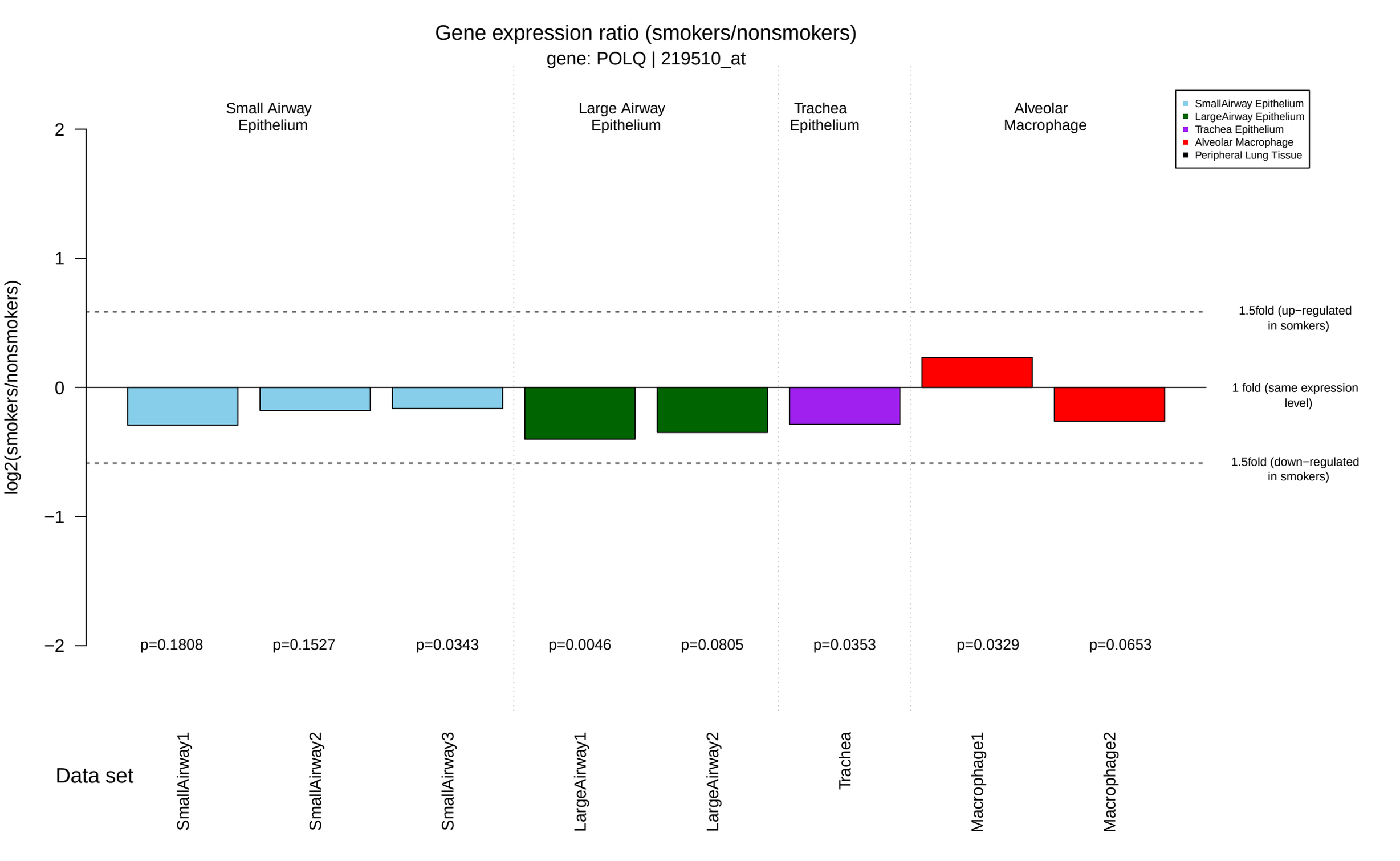
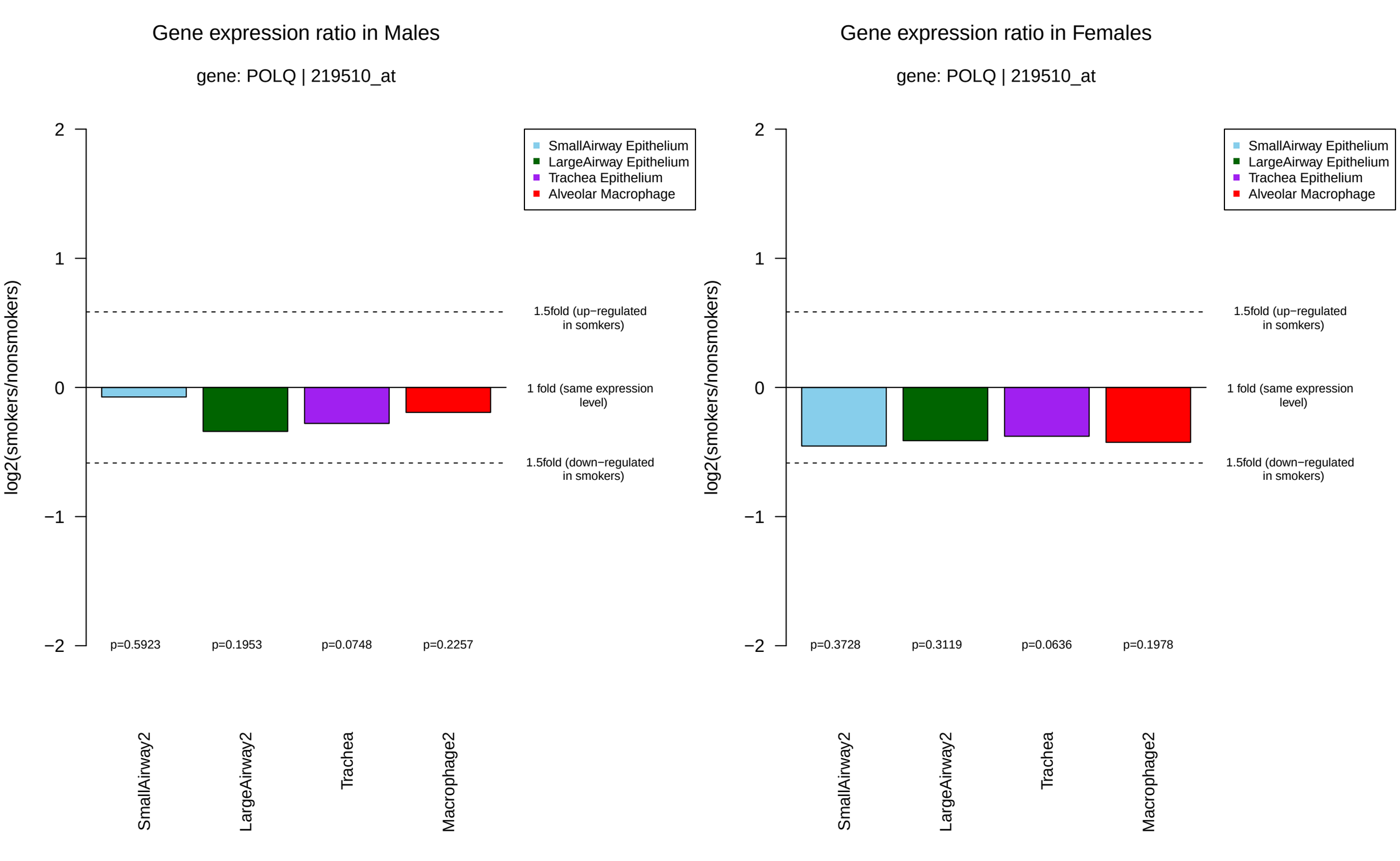
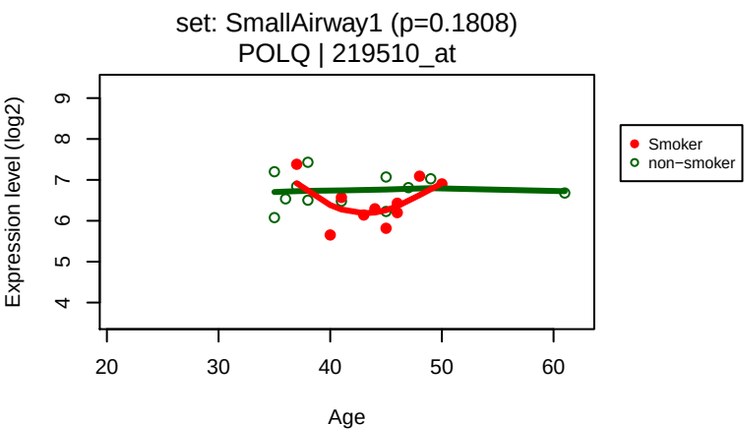
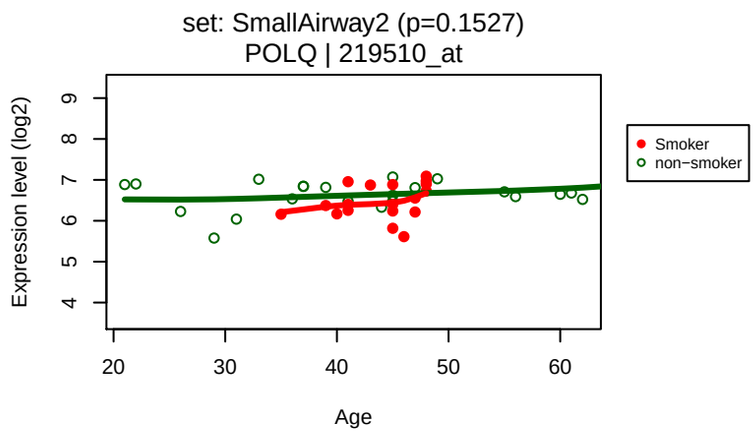
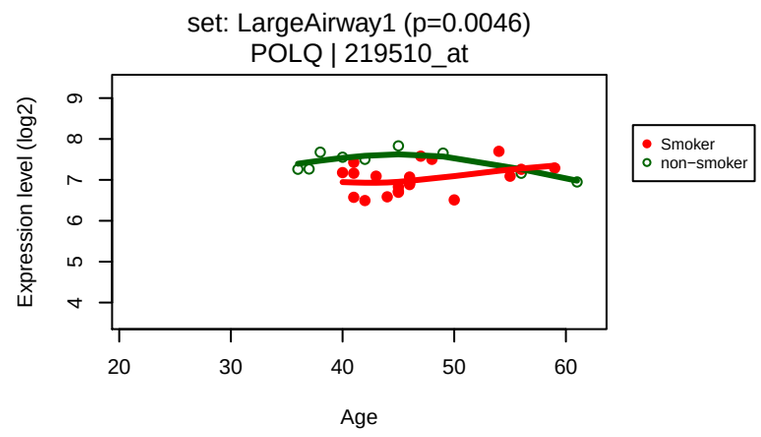

Gene expression of smokers and nonsmokers
gene: POLQ | 219510_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.1808
p=0.1527
p=0.0343
p=0.0046
p=0.0805
p=0.0353
p=0.0329
p=0.0653
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

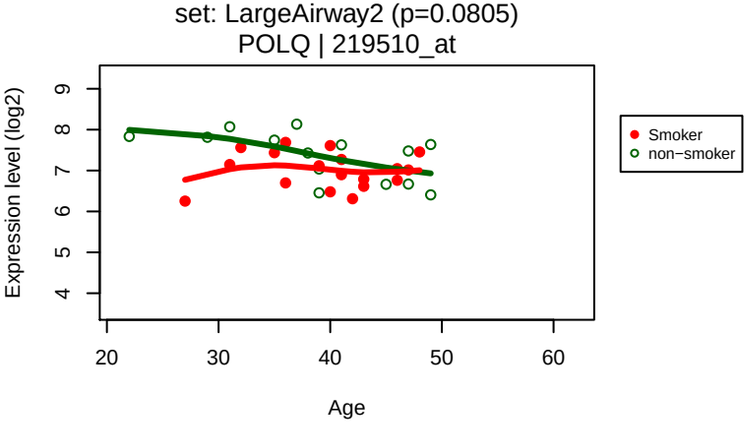

Gene expression ratio (smokers/nonsmokers)
gene: POLQ | 219510_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1808
p=0.1527
p=0.0343
p=0.0046
p=0.0805
p=0.0353
p=0.0329
p=0.0653
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

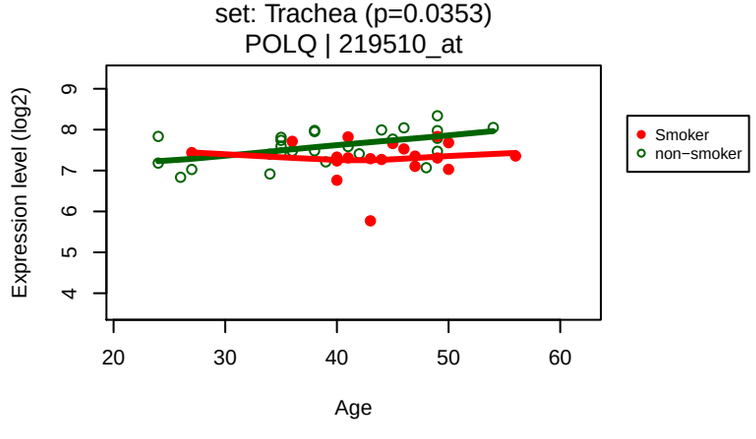

Gene expression ratio in Males
gene: POLQ | 219510_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5923
p=0.1953
p=0.0748
p=0.2257
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: POLQ | 219510_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3728
p=0.3119
p=0.0636
p=0.1978
SmallAirway2
LargeAirway2
Trachea
Macrophage2

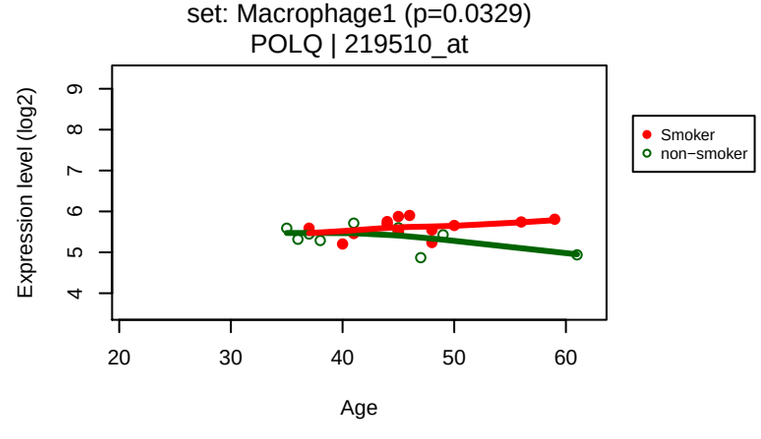

set: SmallAirway1 (p=0.1808)
POLQ | 219510_at
Expression level (log2)
Age
Smoker
non−smoker

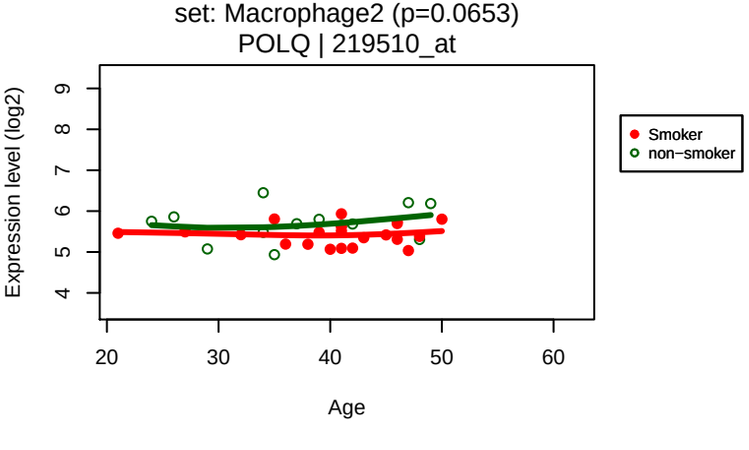

set: SmallAirway2 (p=0.1527)
POLQ | 219510_at
Expression level (log2)
Age
Smoker
non−smoker

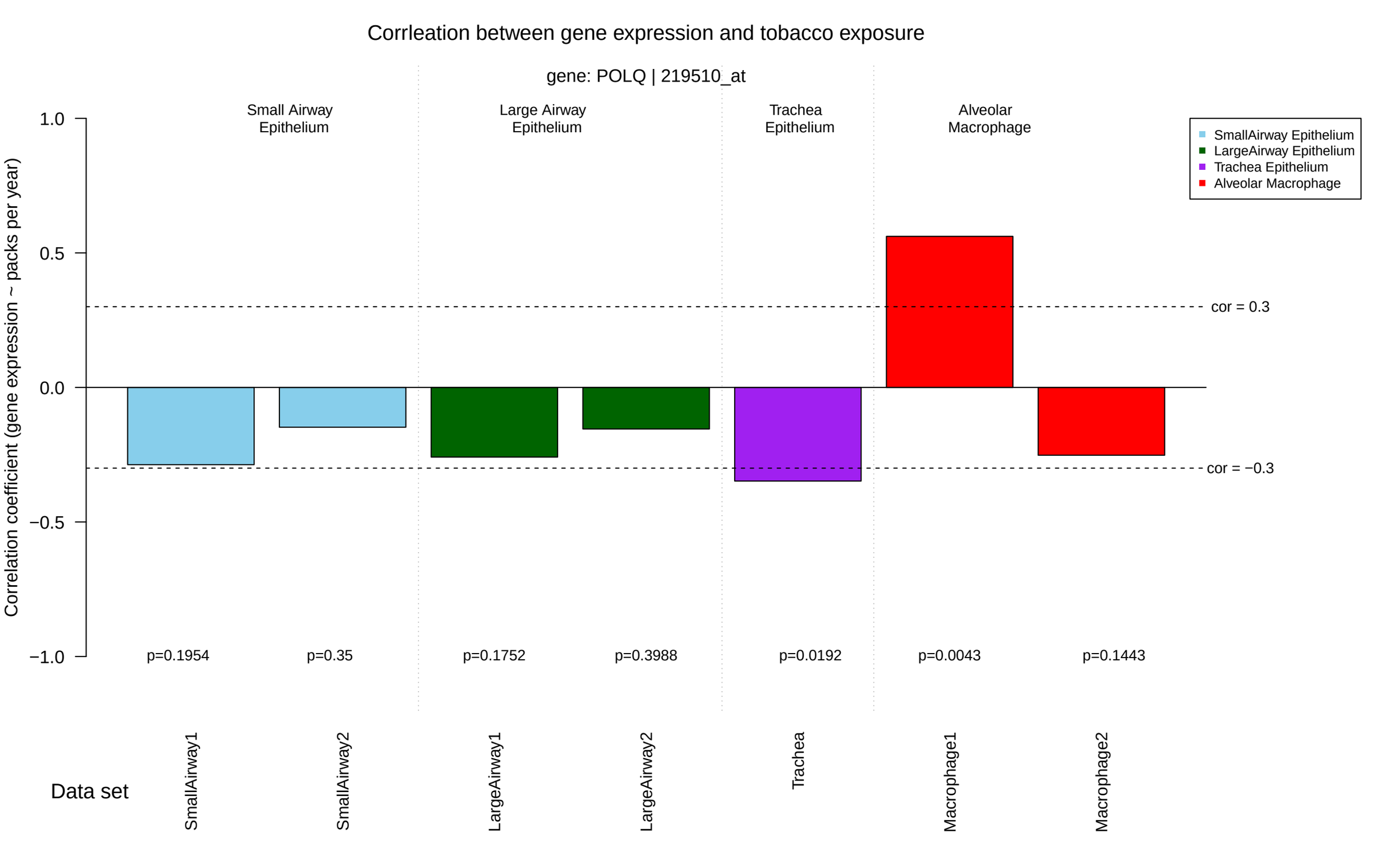

set: LargeAirway1 (p=0.0046)
POLQ | 219510_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0805)
POLQ | 219510_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0353)
POLQ | 219510_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0329)
POLQ | 219510_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.0653)
POLQ | 219510_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: POLQ | 219510_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.1954
p=0.35
p=0.1752
p=0.3988
p=0.0192
p=0.0043
p=0.1443
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2